Max Martinez

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Education

New Jersey Institute of Technology — Newark, NJ
B.S. in Mechanical Engineering Technology — *Expected May 2028*
Organizations: ASME, IEEE, NJIT Prosthetics Club

Experience

Robotics Engineering Intern — Mar. 2026 – Sept. 2026
Human Friendly Robotics ($600k Pre-seed) — *Marlboro, NJ*

- Led **3** interns through design, build, and testing of "Trey," an autonomous data center cable-management robot whose Ethernet test switch verifies up to **8 cables** at once
- Designed Fusion 360 assemblies with linear actuators, motors, hinge clamps, and custom mounting brackets
- Maintained a **6-printer** FDM fleet and tuned print settings and material (PETG), cutting failed prints from **2–5 to 1–2 per day** (∼**50%** reduction)
- Integrated motors, wheel encoders, and RoboClaw controllers with a Teensy 4.1 and Raspberry Pi for motor control and odometry navigation
- Assembled and wired the chassis for "Tyler," a **50-lb** autonomous flooring robot (NVIDIA Jetson) rated for ±**1 mm** tile placement; calibrated placement and added an IMU to stabilize tile pickup

Secretary & Animal Prosthetics Project Lead — Feb. 2026 – Present
NJIT Prosthetics Club — *Newark, NJ*

- Co-leading the design of a custom prosthetic for a three-legged dog
- Review member CAD designs and standardize parts to move teams from prototypes to full assemblies
- Optimized FDM print settings (**45%** infill, **5** wall loops) to improve strength and impact resistance
- Manage communications and documentation between project leads and **50+** active members

Projects

NeuroMaker and BCI Prosthetic Hand | *C++, Arduino* — Feb. 2026 – Apr. 2026

- Assembled a multi-articulated prosthetic hand driven by high-torque servo motors and custom linkages
- Programmed C++ firmware on Arduino to convert EEG attention metrics into distinct gestures
- Calibrated attention-threshold ranges (**0**–**100** scale) and signal filters to reduce false activations

Groundwork | *Dell x NVIDIA AI Hackathon* — Sept. 2026

- Placed **2nd of 40 teams**; helped build a predictive maintenance prototype that flags mechanical fault signatures (e.g., bearing inner-race faults) from vibration sensor data
- Worked in a **4-person** cross-university team (ECE, CS, CE) to deliver a working prototype

Autonomous 3D Print Removal Arm | *Viam Robotics Hackathon, IEEE NJIT* — Nov. 2026

- Prototyped an autonomous print-removal system on a **UFactory xArm 6** with a gripper and arm-mounted **Intel RealSense** depth camera, integrated through Viam, to clear finished prints
- Built and labeled a custom image dataset and trained a **YOLO** detection model to classify failed vs. good prints
- Designed a closed-loop workflow that verifies a cleared bed and restarts jobs, with future G-code analysis for early failure alerts

Technical Skills

CAD and Design: Fusion 360, SolidWorks, Inventor, Onshape, AutoCAD, KiCad
Manufacturing and Prototyping: FDM 3D printing, print-fleet maintenance, assembly, QA inspection
Electronics and Controls: NVIDIA Jetson, Teensy 4.1, Arduino, Raspberry Pi, ESP-32, RoboClaw Controllers, Encoders, IMU Sensors
Programming: C++, Python, MATLAB